

ESQUEL, Argentina

WMO: 878030

Lat: 42.9058S

Lon: 71.1450W

Elev: 2615

StdP: 13.36

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.4	22.3	12.4	11.4	32.7	15.9	13.5	35.7	43.6	46.6	35.1	45.3	3.1	20	0.742

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.3	82.1	58.5	78.0	56.7	74.5	55.0	59.7	78.2	57.9	75.1	56.2	72.1	11.1	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
51.3	61.7	65.2	49.5	57.6	64.3	47.6	53.8	62.7	27.3	78.9	26.0	75.3	24.9	72.1	68.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 38.8	32.1	28.0	DB	11.7	88.1	5.5	2.9	7.7	90.2	4.5	91.9	1.4	93.6	-2.7	95.7	(4)
(5)			WB	10.9	63.1	5.4	2.3	7.0	64.8	3.8	66.1	0.8	67.4	-3.1	69.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	47.9	59.3	59.1	54.7	47.9	42.4	37.5	36.3	39.0	42.8	47.5	52.0	56.6	
	DBStd	9.97	6.36	6.57	5.76	5.66	6.07	6.43	6.81	5.32	5.15	5.13	5.53	6.05	
	HDD50	1906	5	5	19	103	245	377	425	343	222	111	39	12	
	HDD65	6304	198	185	321	512	701	825	890	806	667	542	390	267	
	CDD50	1132	293	261	166	42	9	1	1	2	5	34	100	218	
	CDD65	53	21	21	2	0	0	0	0	0	0	0	1	8	
	CDH74	868	298	322	67	0	0	0	0	0	0	3	35	143	
	CDH80	190	66	85	9	0	0	0	0	0	0	0	4	26	
Wind	WSAvg	11.6	14.9	12.1	11.1	10.4	9.2	9.4	9.1	9.8	10.7	12.6	14.6	15.1	
Precipitation	PrecAvg	18.6	0.9	0.7	0.8	1.6	2.8	3.2	2.7	2.2	1.3	1.1	0.8	0.9	
	PrecMax	27.8	4.8	2.4	2.8	6.5	6.9	8.4	7.4	6.3	4.6	4.1	2.3	4.0	
	PrecMin	10.3	0.0	0.0	0.1	0.1	0.1	0.4	0.1	0.5	0.0	0.0	0.0	0.0	
	PrecStd	4.8	0.9	0.7	0.6	1.3	1.9	1.9	1.7	1.3	1.0	0.9	0.6	0.9	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	86.1	87.5	80.7	71.2	62.3	56.4	56.1	59.1	64.9	72.0	78.9	83.5	
		MCWB	60.5	60.9	58.2	54.1	50.2	46.2	44.9	47.0	47.7	51.4	55.5	59.8	
	2%	DB	81.3	82.4	75.1	65.7	58.7	52.3	51.5	54.1	59.8	66.6	72.5	77.8	
		MCWB	58.1	58.4	56.2	51.6	48.5	44.0	42.9	44.3	45.7	49.4	52.8	56.3	
	5%	DB	77.0	77.6	71.3	62.2	55.4	49.8	48.5	51.1	56.1	62.3	67.9	73.7	
		MCWB	56.5	56.6	54.2	49.5	46.7	42.3	41.3	42.5	44.0	47.2	50.7	54.0	
	10%	DB	73.0	73.4	67.0	59.1	52.4	47.1	45.9	48.2	52.8	58.6	63.9	69.6	
		MCWB	54.6	55.0	52.6	48.1	44.6	40.4	39.5	40.6	42.3	45.7	48.9	52.5	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.8	62.7	60.0	55.6	52.0	47.8	46.4	48.8	49.3	53.6	56.7	61.6
			MCDB	83.3	84.2	77.3	67.8	59.6	54.6	52.6	57.1	61.8	68.4	75.7	81.0
2%		WB	59.3	59.5	57.5	52.8	49.5	44.9	43.7	45.4	46.8	50.4	53.9	57.5	
		MCDB	78.2	77.9	72.3	62.5	56.9	51.1	50.0	52.2	57.6	63.9	69.7	75.5	
5%		WB	57.2	57.8	55.4	50.8	47.3	42.9	42.0	43.3	44.8	48.4	51.6	55.2	
		MCDB	74.6	75.1	69.1	59.8	54.1	48.7	47.5	49.6	54.1	60.6	65.9	71.3	
10%		WB	55.4	55.8	53.4	49.0	45.4	41.0	40.0	41.2	43.0	46.4	49.5	53.2	
		MCDB	70.7	71.4	65.5	57.8	51.5	45.9	45.0	47.1	51.4	56.8	62.4	68.0	
Mean Daily Temperature Range	5% DB	MDBR	20.3	21.4	20.2	18.2	15.0	12.5	13.2	14.5	17.4	18.5	19.0	19.3	
		MCDBR	29.0	29.8	28.6	24.5	18.7	15.0	15.8	18.9	23.8	27.2	27.3	28.4	
		MCWBR	12.2	12.6	13.8	13.4	11.5	9.8	9.8	11.2	12.8	13.7	13.1	12.5	
	5% WB	MCDBR	25.8	26.8	25.2	20.9	16.6	14.5	14.7	16.6	20.5	24.4	24.1	25.6	
		MCWBR	11.9	12.2	13.0	12.2	10.8	10.0	9.6	10.6	11.7	13.1	12.3	12.2	
Clear-Sky Solar Irradiance	taub	0.298	0.302	0.291	0.282	0.275	0.264	0.263	0.277	0.295	0.297	0.296	0.302		
	taud	2.587	2.568	2.601	2.616	2.593	2.576	2.562	2.566	2.522	2.537	2.561	2.552		
	Ebn at Noon	327	317	306	287	266	259	267	285	300	315	325	327		
	Edh at Noon	32	31	27	23	19	18	19	23	29	32	33	34		
All-Sky Solar Radiation	RadAvg	2423	2101	1527	992	583	429	503	749	1186	1663	2153	2412		
	RadStd	123	124	103	81	68	51	47	72	105	123	126	167		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	+1.53	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	+1.55	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page